

By Theorem 2, the system of linear equations corresponding to the augmented matrix is consistent if and only if the echelon form of the matrix does not contain any row of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$, where $b \neq 0$. It has a unique solution if and only if all the columns of A are pivot columns.

Suppose, by the row reduction algorithm we convert the Augmented matrix $[a_1 \ a_2 \ a_3 \ b]$ into its rref form. By Theorem 1, this matrix is unique. A matrix converted to rref form must satisfy the following conditions:

1. Every non-zero row must be above all zero rows.
2. For each row, its leading entry must be in a column to the right of the columns of the leading entries of all rows above it.
3. For each column containing a leading entry, all positions below the leading entry position must contain only 0.
4. All leading entries must be 1.
5. For each column containing a leading entry, all positions above the leading entry position must contain only 0.

Since the equation $Ax=b$ has a unique solution, each of the columns a_1, a_2 and a_3 must have a pivot in it. $\therefore$ In rref form, the matrix A will be $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Of course, the 4th entry of b must be 0 in rref form due to the consistency of the system of equations, (Theorem 2). $\square$

35) Let A be a 3×4 matrix, let y_1 and y_2 be vectors in $\mathbb{R}^3$, and let $w = y_1 + y_2$. Suppose $y_1 = Ax_1$ and $y_2 = Ax_2$ for some vectors x_1 and x_2 in $\mathbb{R}^4$. What fact allows you to conclude that the system $Ax=w$ is consistent? (Note: x_1 and x_2 denote vectors, not scalar entries in vectors.)

Solution: $w = y_1 + y_2 = Ax_1 + Ax_2 = A(x_1 + x_2)$ (Theorem 5(a))
 $= Ax$ (By Vector addition)

So the vector $x = x_1 + x_2$ is a solution of $Ax = w$. $\square$